
Z7-LITE Reference Manual

[中文]

Development Environment:

Vivado 2018.3 is from Xilinx website

<https://www.xilinx.com>

WeChat Public Number: 

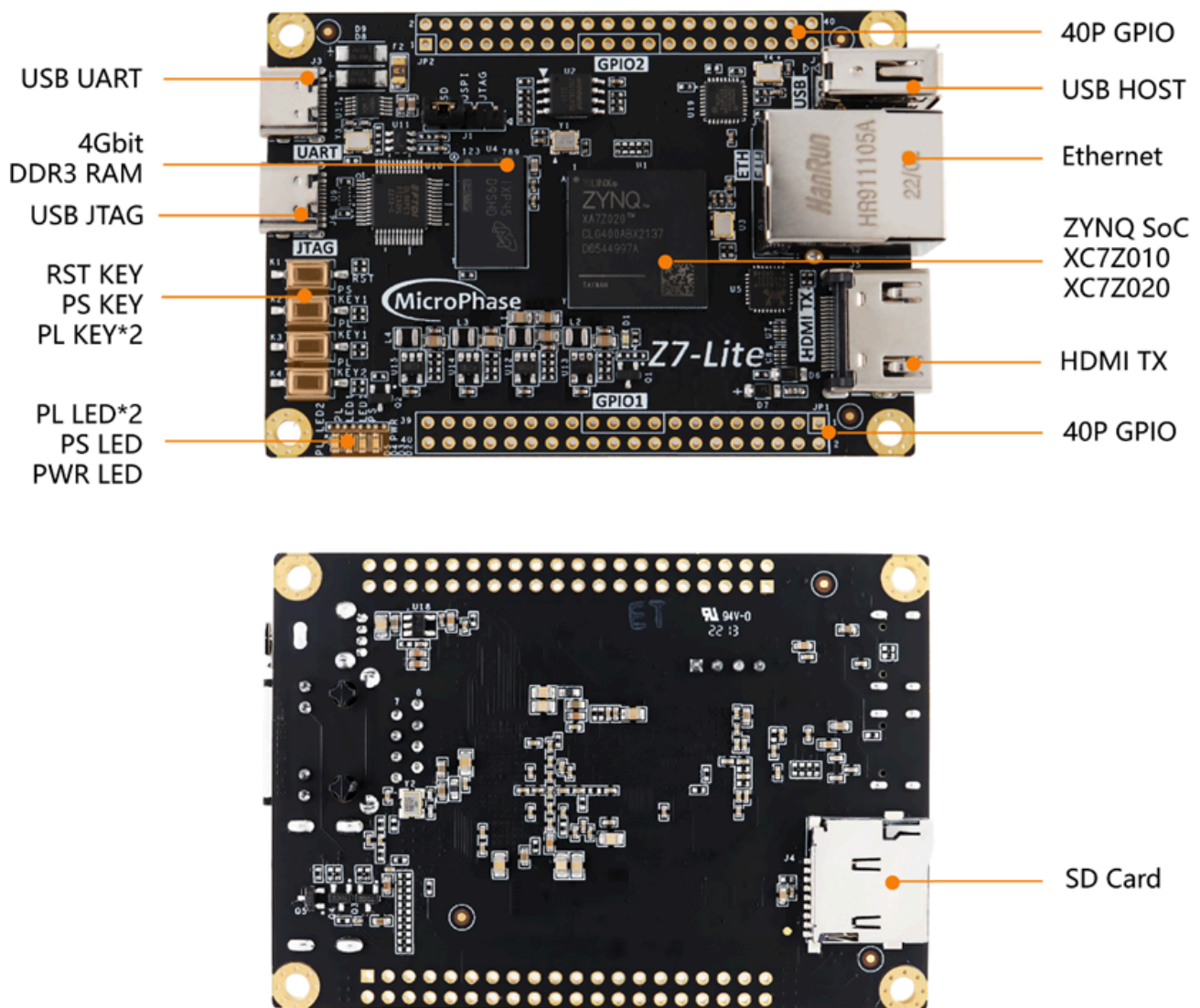


●1. Overview

Z7 Lite is a commercial-grade development board based on Xilinx Zynq-7000 SoCs (XC7Z010 or XC7Z020) with up to 4Gb of DDR3/L SDRAM, 128MB of SPI flash memory, Gigabit Ethernet PHY transceiver, a USB PHY transceiver and an easy way to support power. With customizable variants available upon request, customised requirements may be subject to minimum order quantities, please contact our sales team for more information: sales@microphase.cn.

It is suitable for software verification and hardware development, helping to accelerate project progress.

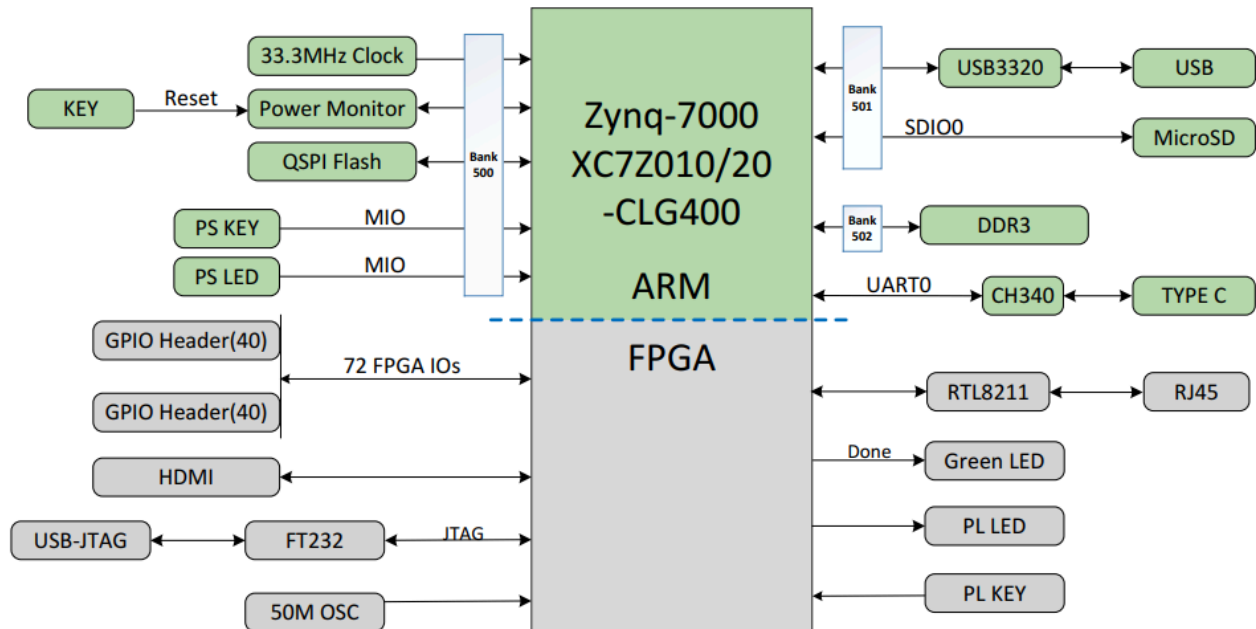
○Board Layout



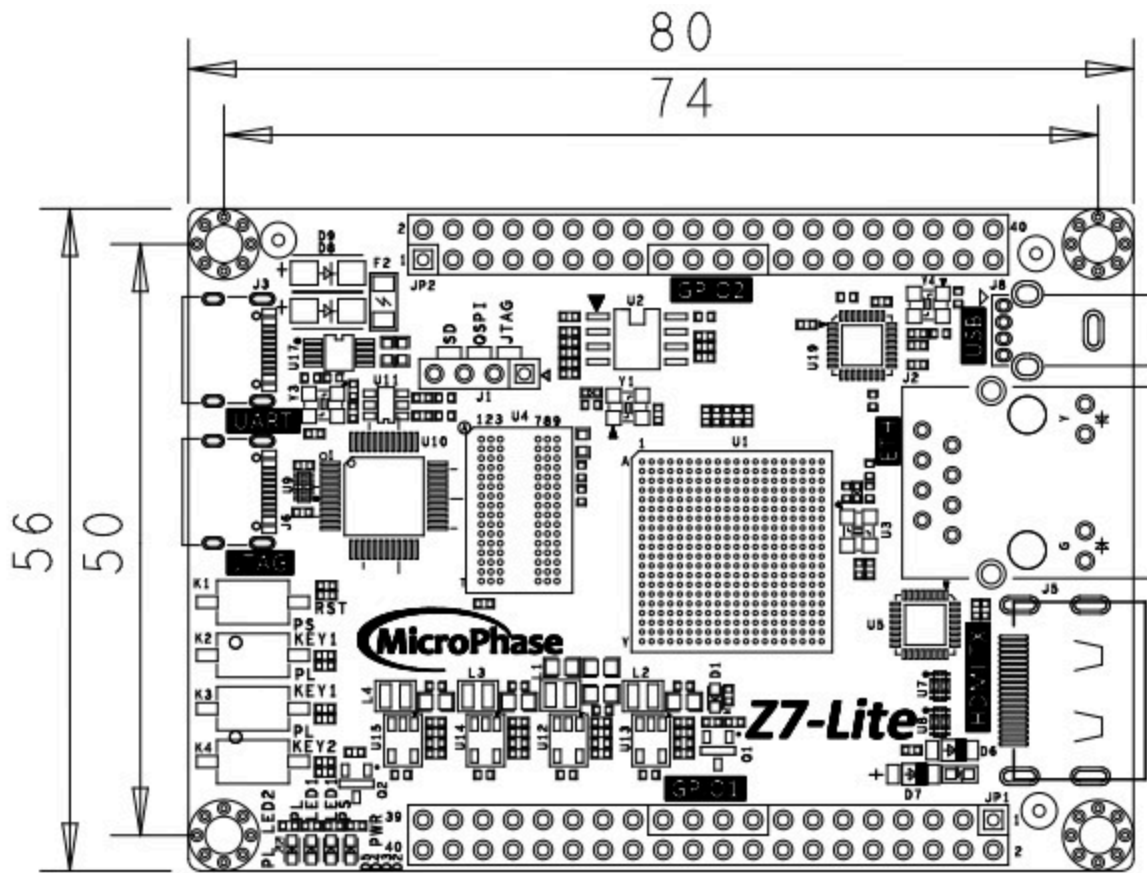
○Key Features

- Xilinx Zynq™ XC7Z010-1CLG400C (7010 Version Only),
Xilinx Zynq™ XC7Z020-1CLG400C (7020 Version Only).
- DDR3: 1 4Gbit,DDR3.(MT41J256M16HA-125:K or equivalent)
- KEY: 3 user's keys, 1 PS control, 2 PL control.
- LED: 3 user's LEDs, 1 PS control, 2 PL control.
- GPIO: 2 40 pin expansion ports (2.54mm pitch), 72 IOs of 3.3V voltage.
- USB JTAG: 1 on board JTAG circuit, can debug and download the ZYNQ system through the USB cable.
- HDMI: 1 HDMI video output interface, can realize 1080P video image.
- PHY ETH: 1 10/100 Ethernet RJ45 interface, which can be use for Ethernet data exchange or other application.
- USB Host: 1 USB Host.
- SD Card: 1 SD Card is provided to store the operating system image and file system.
- USB UART: 1 USB UART interface, using for serial communication with PC.
- CLOCK: 1 33.33Mhz active crystal oscillator provides a stable clock for the PS system.
1 50Mhz active crystal oscillator provides additional clocks for the PL logic.

○Block diagram



○Mechanical Spec



●2. Function Resources

○FPGA

- 667 MHz dual-core Cortex-A9 processor
- DDR3L memory controller with 8 DMA channels and 4
- High Performance AXI3 Slave ports
- High-bandwidth peripheral controllers: 1G Ethernet, USB 2.0, SDIO
- Low-bandwidth peripheral controllers: SPI, UART, CAN, I2C
- Programmable from JTAG, Quad-SPI flash, and microSD card
- Programmable logic equivalent to Artix-7 FPGA

LUTs: 17,600 (7010)

53,200(7020)

DSP Slices: 80 (7010)

220 (7020)

Logic Cells: 28K (7010)

85K(7020)

Flip-Flops: 35,200 (7010)

106,400 (7020)

Total Block RAM: 2.1Mb (7010)

4.9Mb (7020)

- Analog Mixed Signal (AMS) / XADC: 2x 12 bit, MSPS ADCs with up to 17 Differential Inputs
- Security: AES & SHA 256b Decryption & Authentication for Secure Programmable Logic Config

○DDR3

One on-board 16bit 512M DDR3/L SDRAM chips.

Position	Model	Capacity	Factory
U4	MT41K256M16	256M x 16bit	Micron

Signal Name	PIN Number	Signal Name	PIN Number
PS_DDR3_A0	N2	PS_DDR3_D0	C3
PS_DDR3_A1	K2	PS_DDR3_D1	B3
PS_DDR3_A2	M3	PS_DDR3_D2	A2
PS_DDR3_A3	K3	PS_DDR3_D3	A4
PS_DDR3_A4	M4	PS_DDR3_D4	D3
PS_DDR3_A5	L1	PS_DDR3_D5	D1
PS_DDR3_A6	L4	PS_DDR3_D6	C1
PS_DDR3_A7	K4	PS_DDR3_D7	E1
PS_DDR3_A8	K1	PS_DDR3_D8	E2
PS_DDR3_A9	J4	PS_DDR3_D9	E3
PS_DDR3_A10	F5	PS_DDR3_D10	G3
PS_DDR3_A11	G4	PS_DDR3_D11	H3
PS_DDR3_A12	E4	PS_DDR3_D12	J3
PS_DDR3_A13	D4	PS_DDR3_D13	H2
PS_DDR3_A14	F4	PS_DDR3_D14	H1
PS_DDR3_BA0	L5	PS_DDR3_D15	J1
PS_DDR3_BA1	R4	PS_DDR3_DQS_N0	B2

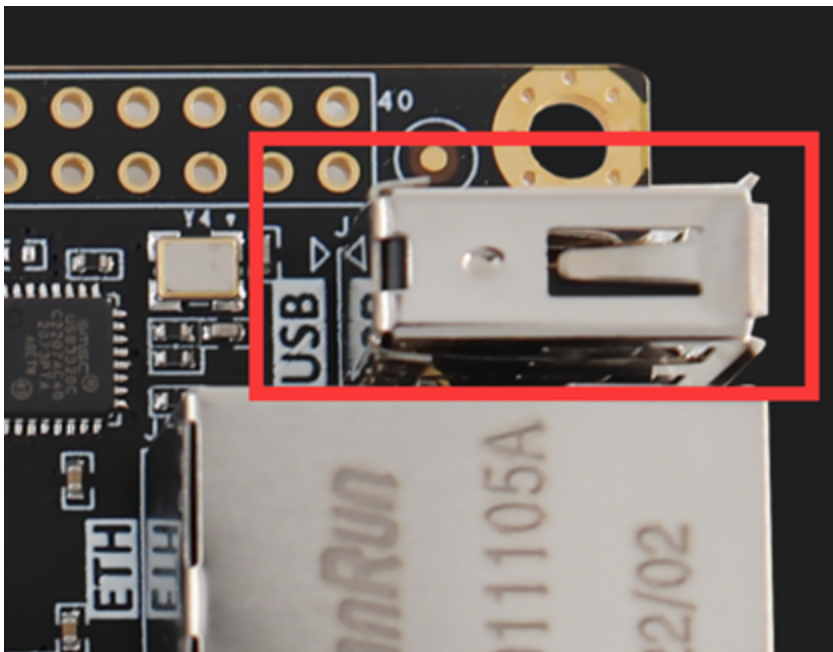
Signal Name	PIN Number	Signal Name	PIN Number
PS_DDR3_BA2	J5	PS_DDR3_DQS_N1	F2
PS_DDR3_NCAS	P5	PS_DDR3_DQS_P0	C2
PS_DDR3_CKE	N3	PS_DDR3_DQS_P1	G2
PS_DDR3_CLK_N	M2	PS_DDR3_NRST	B4
PS_DDR3_CLK_P	L2	PS_DDR3_ODT	N5
PS_DDR3_NCS	N1	PS_DDR3_NRAS	P4
PS_DDR3_DM0	A1	PS_DDR3_NWE	M5
PS_DDR3_DM1	F1		

○Giga ETH

The RTL8201F chip supports 10/100M network transfer rate and communicates with the MAC layer of the Zynq7000 PS system via the RGMII interface. It supports MDI/MDX adaptation, multiple speed adaptation, master/slave adaptation and MDIO bus support for PHY register management.

○USB Host

The USB2.0 transceiver on-board is the USB3320C-EZK, which is supports the ULPI standard interface. It's connected with ZYNQ's bus interface to enable high-speed USB2.0 Host mode data communications.



○USB UART

A USB to UART chip, the CH340, is provided for user connection to the host PC.

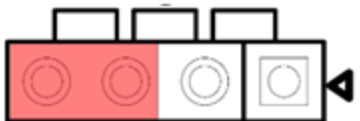
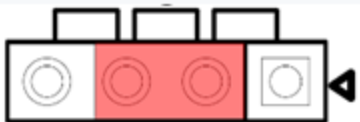
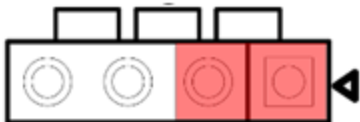
Signal Name	Pin Name	Pin Number	Explain
UART_TX	PS_MIO15_500	C8	UART data output
UART_RX	PS_MIO14_500	C5	UART data input

○USB JTAG

We have incorporated the JTAG download and debug circuitry into the board, enabling users to develop and debug ZYNQ through a USB cable.

○Boot Config

The Z7-Lite startup settings include JTAG, QSPI and SD modes, which are controlled by the MODE input signal from the J1. The user has the option to change mode through the jumper cap. The startup mode is configured as shown in the figure below.

Jumper cap position	Mode
	SD
	QSPI
	JTAG

○Quad-SPI Flash

The onboard 128M Quad-SPI Flash memory W25Q128JVSIQ is used for the storage of the initial FPGA configuration, user applications and data.

Position	Model	Capacity	Factory
U2	W25Q128JVS1Q	128M Byte	Winbond

○Clock

A 33.33 MHz active crystal oscillator provides a stable clock for the PS system, while the other 50 MHz active crystal oscillator provides additional clocks for the PL logic.

Position	Signal Name	Frequency	Pin Number
Y1	PS_CLK_33d3	33.333Mhz	E7
U3	PL_CLK_50M	50Mhz	N18

○Reset

We provide a key (K1) that can be used as a ‘reset’ signal for designs running on FPGA.

○Micro SD

The board provides a Micro SD card interface, allowing users to access the SD card memory. It also provides storage for the ZYNQ chip’s bootloader, Linux operating system kernels, file systems, and other user data files.

○HDMI

One HDMI video output interface can realize 1080P video images.

○LED

We provide three LEDs for user to use. When the corresponding pin of the FPGA is in a low state, the LED is illuminated.

Position	Signal Name	Pin Number	Explain
D3	PS_LED1	E6	PS LED1
D4	PL_LED1	P15	PL LED1
D5	GPIO1_17N	U12	PL LED2

○Key

We provide three keys for user to use. Once the key is pressed, it will be on the low level.

Position	Signal Name	Pin Number	Explain
K2	PS_KEY1	B5	PS KEY1
K3	PL_KEY1	P16	PL KEY1
K4	GPIO1_17P	T12	PL KEY2

○GPIO

This device is equipped with a maximum of 72 user IO pins that can be used for various custom applications. All user IOs are length matched and can be used as differential pairs.

JP1:

Pin	Signal Name	Pin Number	Pin	Signal Name	Pin Number
1	GPIO1_0P	N17	2	GPIO1_0N	P18
3	GPIO1_1P	R16	4	GPIO1_1N	R17
5	GPIO1_2P	T16	6	GPIO1_2N	U17
7	GPIO1_3P	W18	8	GPIO1_3N	W19
9	GPIO1_4P	Y18	10	GPIO1_4N	Y19
11	VCC_5V	-	12	GND	-
13	GPIO1_5P	Y16	14	GPIO1_5N	Y17
15	GPIO1_6P	V17	16	GPIO1_6N	V18
17	GPIO1_7P	W14	18	GPIO1_7N	Y14
19	GPIO1_8P	V16	20	GPIO1_8N	W16
21	GPIO1_9P	T17	22	GPIO1_9N	R18
23	GPIO1_10P	V12	24	GPIO1_10N	W13
25	GPIO1_11P	T14	26	GPIO1_11N	T15
27	GPIO1_12P	T11	28	GPIO1_12N	T
29	VCC_3V3	-	30	GND	-



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latest



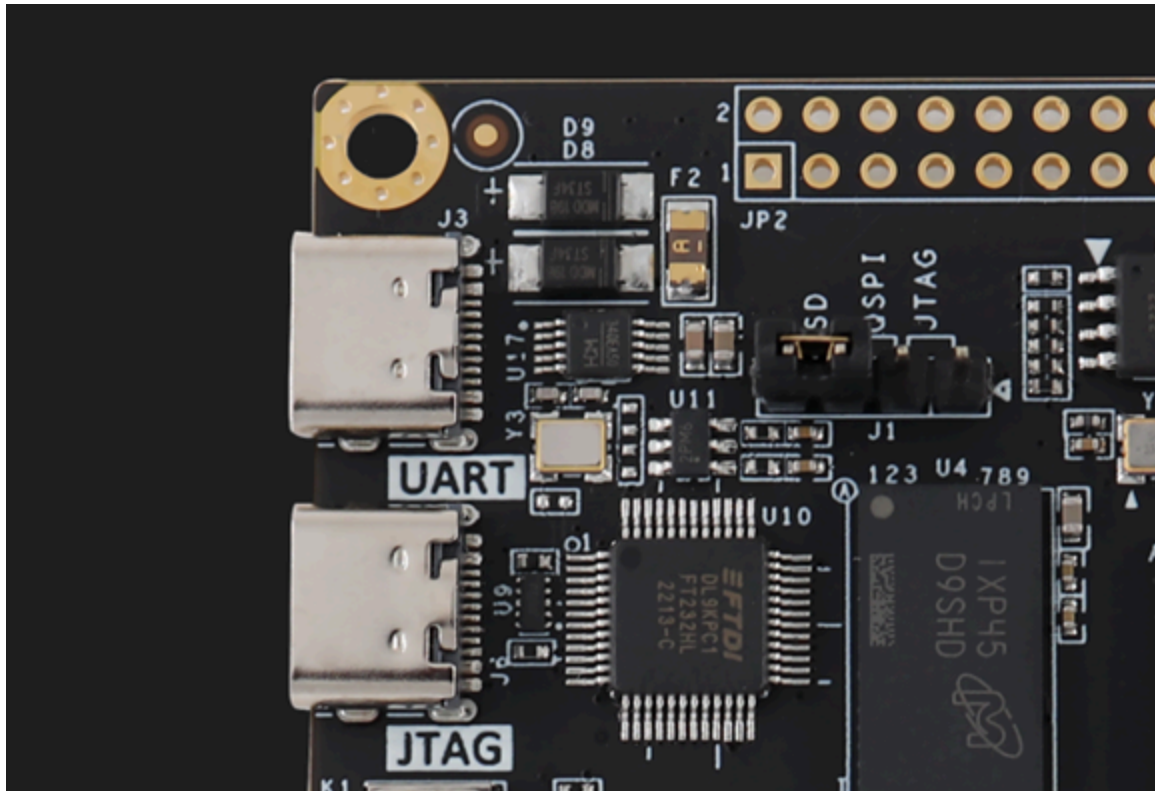
Pin	Signal Name	Pin Number	Pin	Signal Name	Pin Number
31	GPIO1_13P	V15	32	GPIO1_13N	W15
33	GPIO1_14P	P14	34	GPIO1_14N	R14
35	GPIO1_15P	U14	36	GPIO1_15N	U15
37	GPIO1_16P	U13	38	GPIO1_16N	V13
39	GPIO1_17P	T12	40	GPIO1_17N	U12

JP2

Pin	Signal Name	Pin Number	Pin	Signal Name	Pin Number
1	GPIO2_0P	L16	2	GPIO2_0N	L17
3	GPIO2_1P	H15	4	GPIO2_1N	G15
5	GPIO2_2P	F16	6	GPIO2_2N	F17
7	GPIO2_3P	E18	8	GPIO2_3N	E19
9	GPIO2_4P	B19	10	GPIO2_4N	A20
11	VCC_5V	-	12	GND	-
13	GPIO2_5P	D19	14	GPIO2_5N	D20
15	GPIO2_6P	E17	16	GPIO2_6N	D18
17	GPIO2_7P	H16	18	GPIO2_7N	H17
19	GPIO2_8P	G19	20	GPIO2_8N	G20
21	GPIO2_9P	J18	22	GPIO2_9N	H18
23	GPIO2_10P	K16	24	GPIO2_10N	J16
25	GPIO2_11P	C20	26	GPIO2_11N	B20
27	GPIO2_12P	G17	28	GPIO2_12N	G18
29	VCC_3V3	-	30	GND	-
31	GPIO2_13P	L19	32	GPIO2_13N	L20
33	GPIO2_14P	F19	34	GPIO2_14N	F20
35	GPIO2_15P	M19	36	GPIO2_15N	N20
37	GPIO2_16P	K19	38	GPIO2_16N	J20
39	GPIO2_17P	J20	40	GPIO2_17N	H20

○Power

The board uses a +5V supply from USB. The Figure shows the USB can be used to provide power.



●3.Related Documents

- [Z7-Lite_R11 Schematic \(PDF\)](#)
- [Z7-Lite_R11 Dimensions \(PDF\)](#)
- [Z7-Lite_R11 Dimensions source file \(DXF\)](#)